TRƯỜNG ĐẠI HỌC KHOA HỌC TỰ NHIÊN, ĐHQG - HCM
KHOA CÔNG NGHỆ THÔNG TIN

Introduction to Information Retrieval

Interactive Search with Relevance Feedback

Lecturer: Dr. LE Trung Nghia, Dr. NGUYEN Vinh Tiep

Disclaimer

- Disclaimer: This curriculum have been inspired and adapted from materials created by:
  - Richard Szeliski, "Computer Vision: Algorithms and Applications", 2nd edition, 2022. https://szeliski.org/Book/

Relevance Feedback

- After initial retrieval results are presented, allow the user to provide feedback on the relevance of one or more of the retrieved results.
- Use this feedback information to reformulate the query.
- Produce new results based on reformulated query.
- Allows more interactive, multi-pass process

Relevance Feedback Architecture

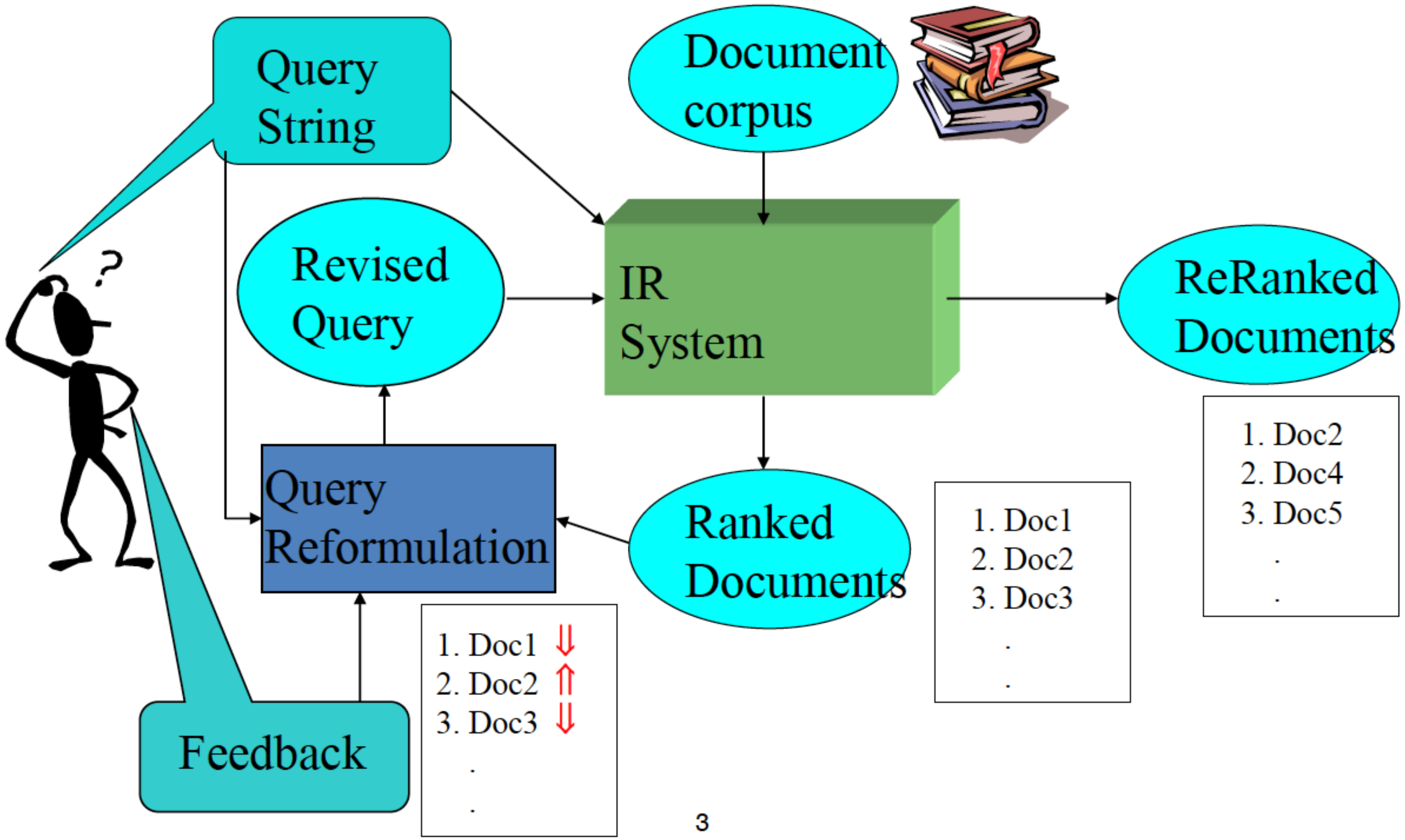
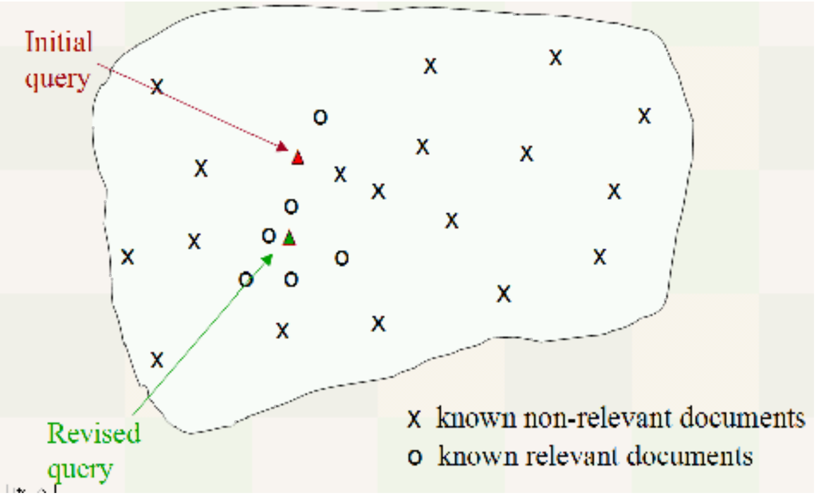


Image Retrieval with Relevance Feedback

- User is incorporatated as part of image retrieval loop
- Advantages:
  - Establish more accurate link between low-level features and high-level concept
  - Better retrieval performance of system
- Steps:
  - 1. System provides an initial set of images according to the query
  - 2. User feeds back positive/negative images
  - 3. System learns from user's feedback
  - 4. System the refines the original query
  - 5. Repeat steps (1)~(4) till user is satisfied



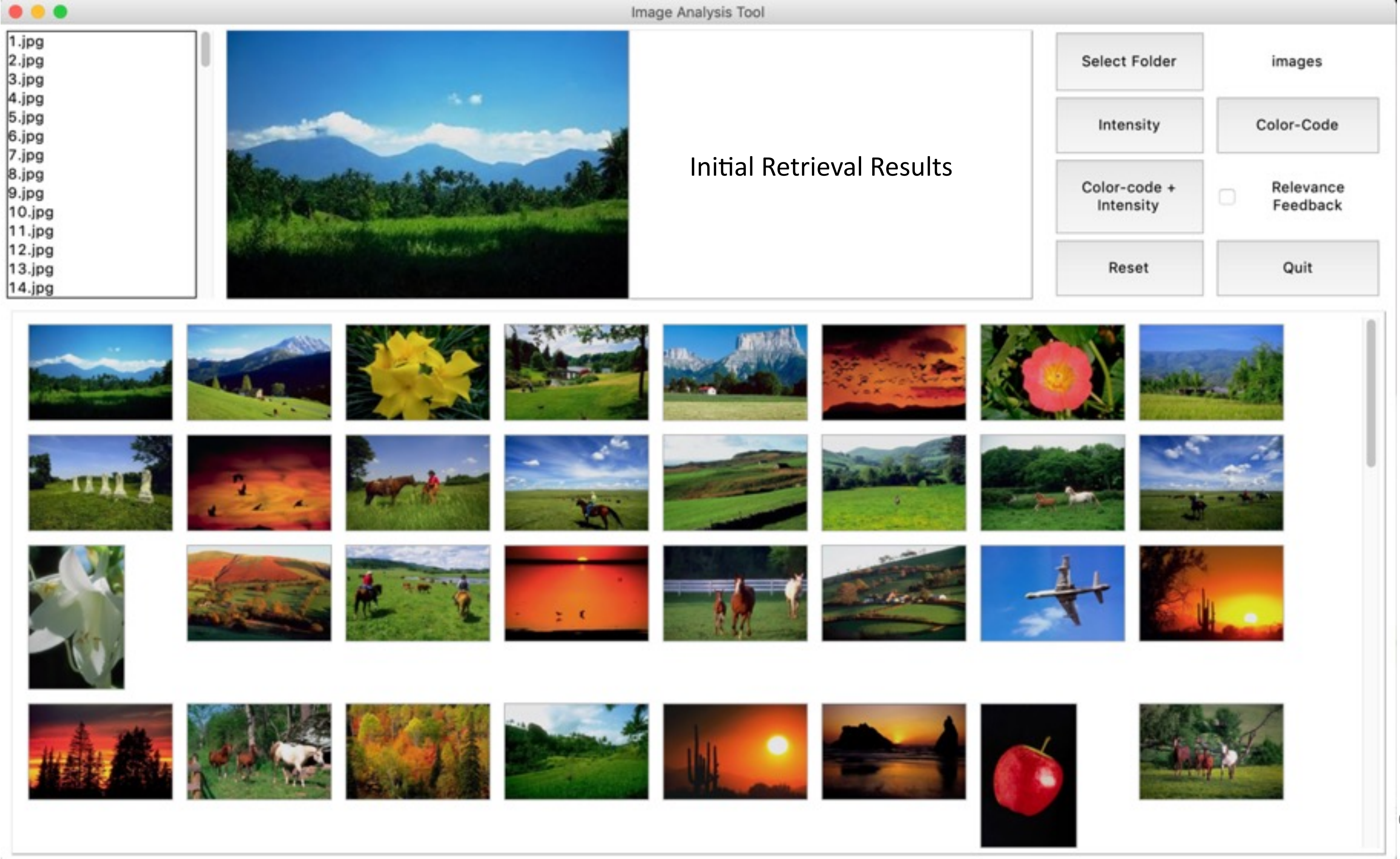


Image Analysis Tool
1.jpg
2.jpg
3.jpg
4.jpg
5.jpg
6.jpg
7.jpg
8.jpg
9.jpg
10.jpg
11.jpg
12.jpg
13.jpg
14.jpg
Initial Retrieval Results
Select Folder
images
Intensity
Color-Code
Color-code + Intensity
Relevance Feedback
Reset
Quit

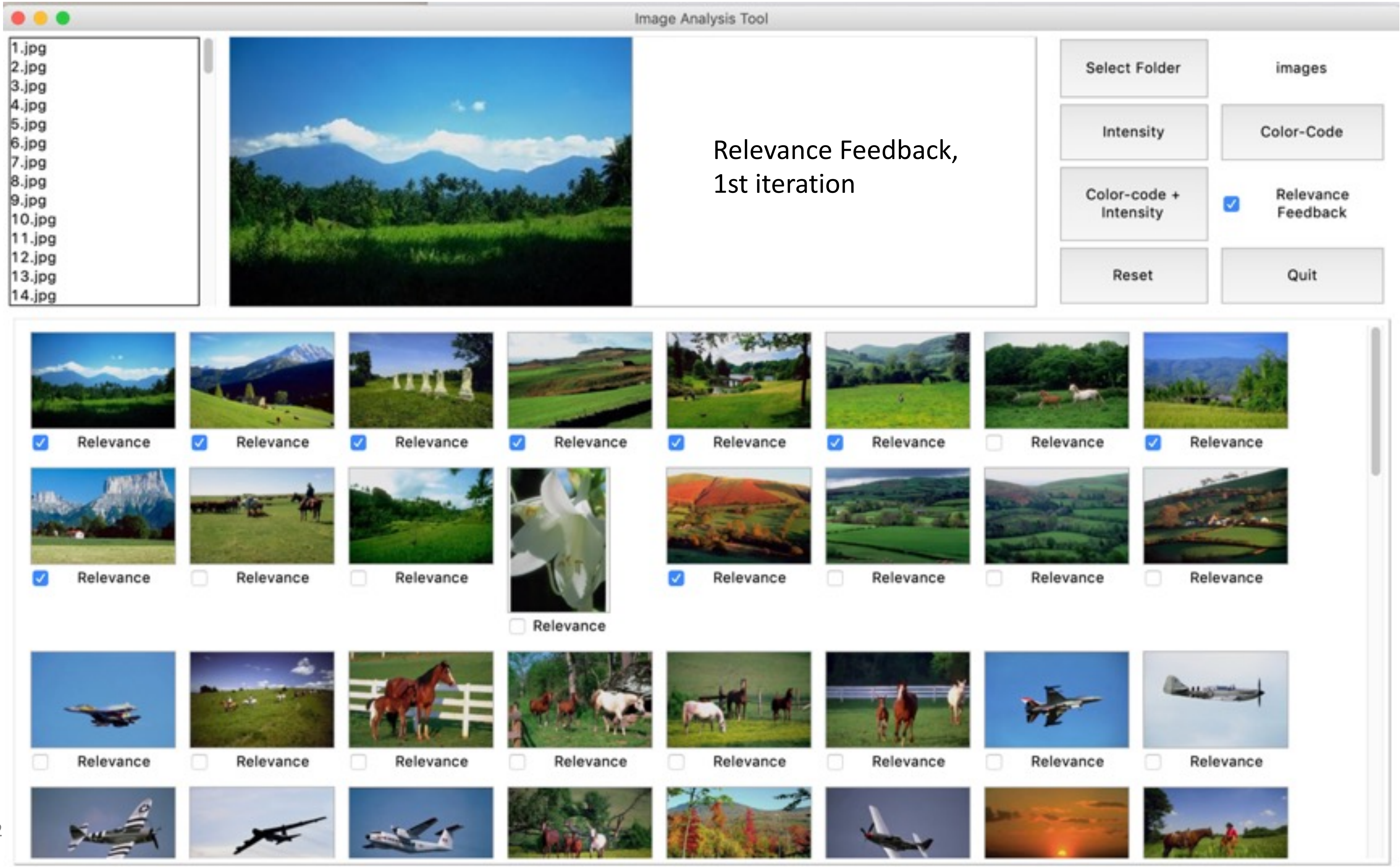
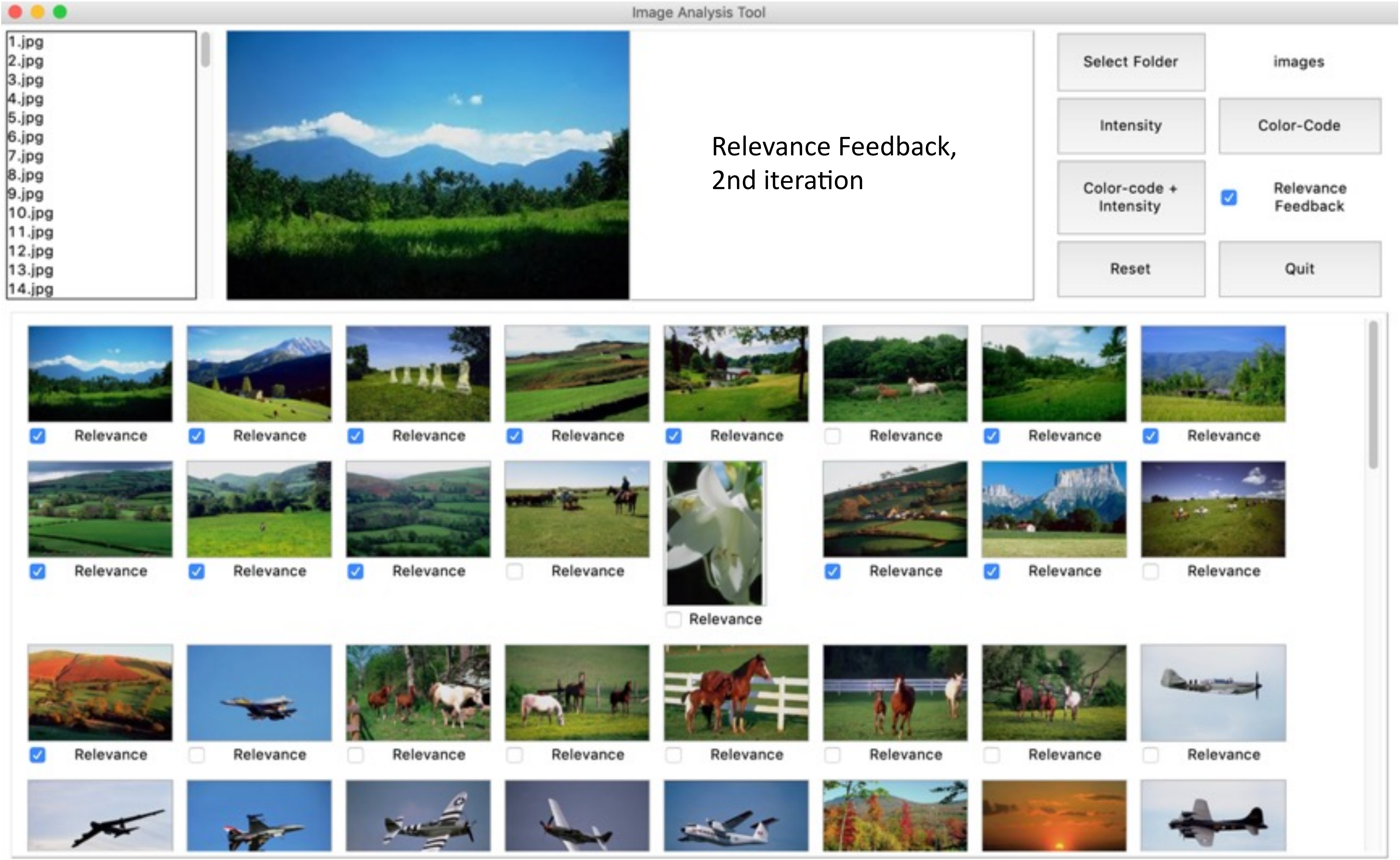
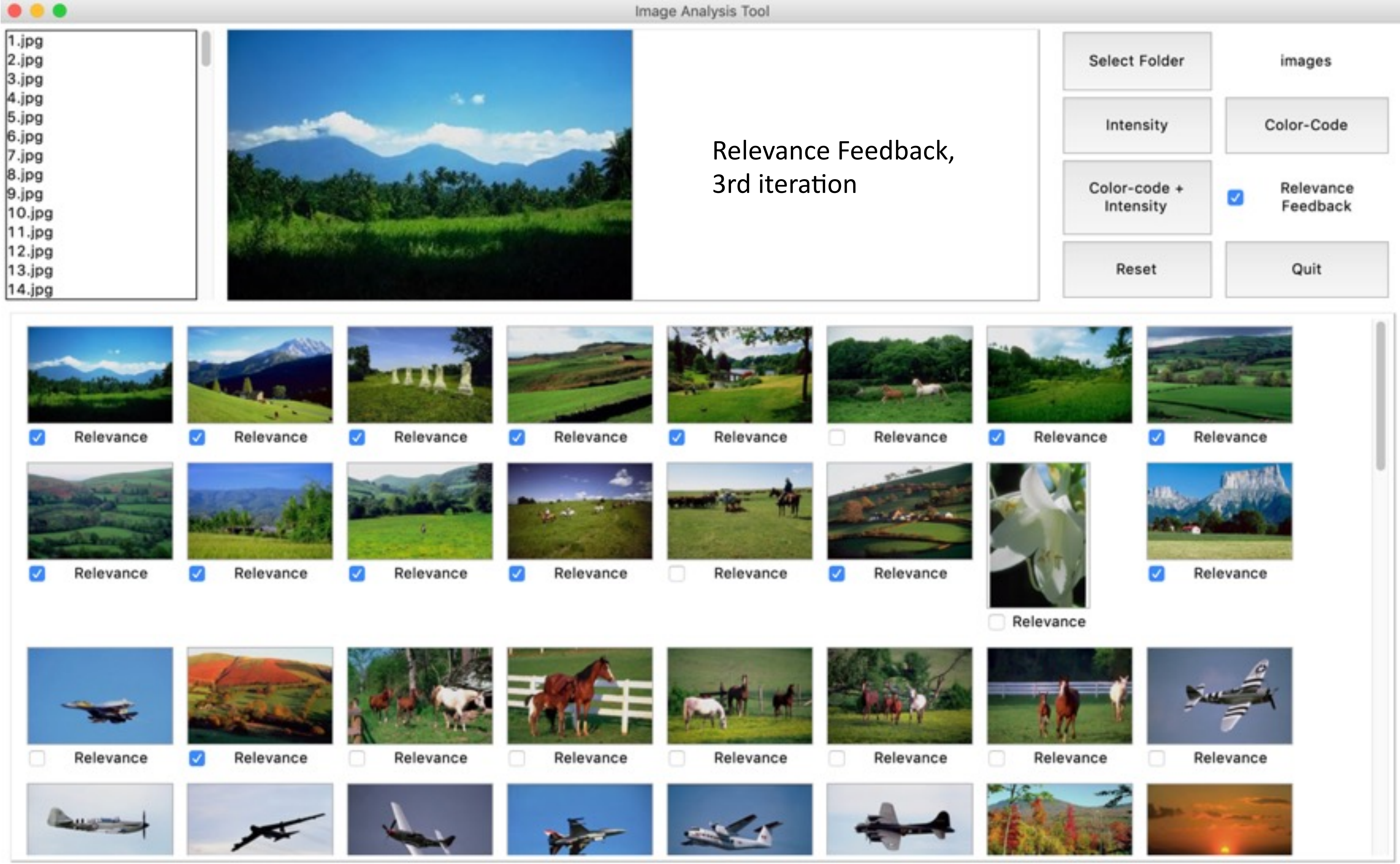


Image Analysis Tool
1.jpg
2.jpg
3.jpg
4.jpg
5.jpg
6.jpg
7.jpg
8.jpg
9.jpg
10.jpg
11.jpg
12.jpg
13.jpg
14.jpg
Relevance Feedback,
1st iteration
Select Folder
images
Intensity
Color-Code
Color-code + Intensity
Relevance Feedback
Reset
Quit
Relevance



Relevance Feedback,
2nd iteration

Relevance Feedback,
3rd iteration

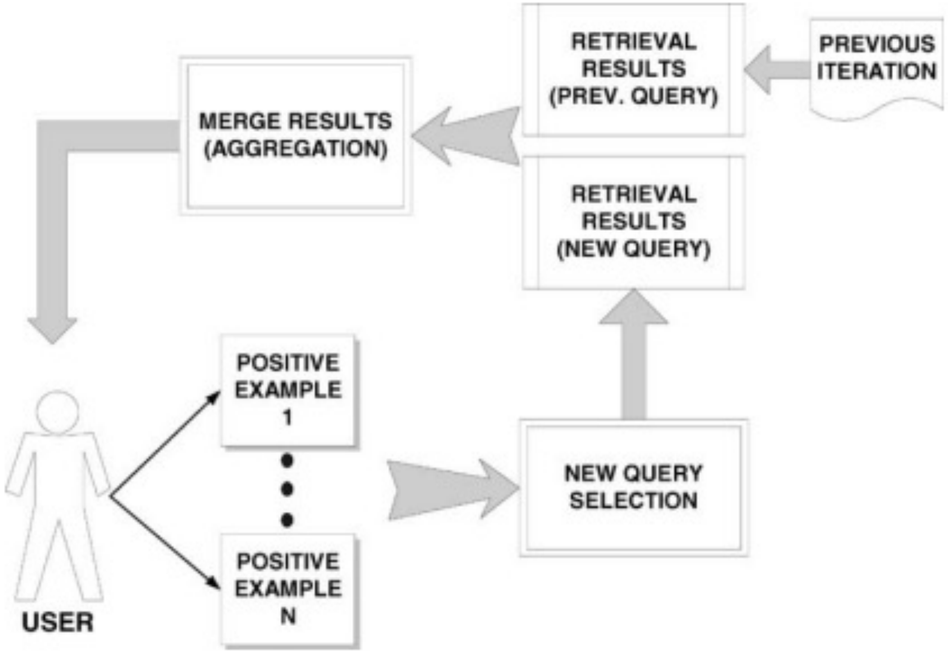


Query Reformulation

- Revise query to account for feedback:
  - Term Reweighting: Increase weight of terms in relevant documents and decrease weight of terms in irrelevant documents
  - Query Expansion: Add new terms to query from relevant documents.

Term Reweighting

- Change query vector using vector algebra.
  - Add the vectors for the relevant documents to the query vector.
  - Subtract the vectors for the irrelevant docs from the query vector.
- This both adds both positive and negatively weighted terms to the query as well as reweighting the initial terms.

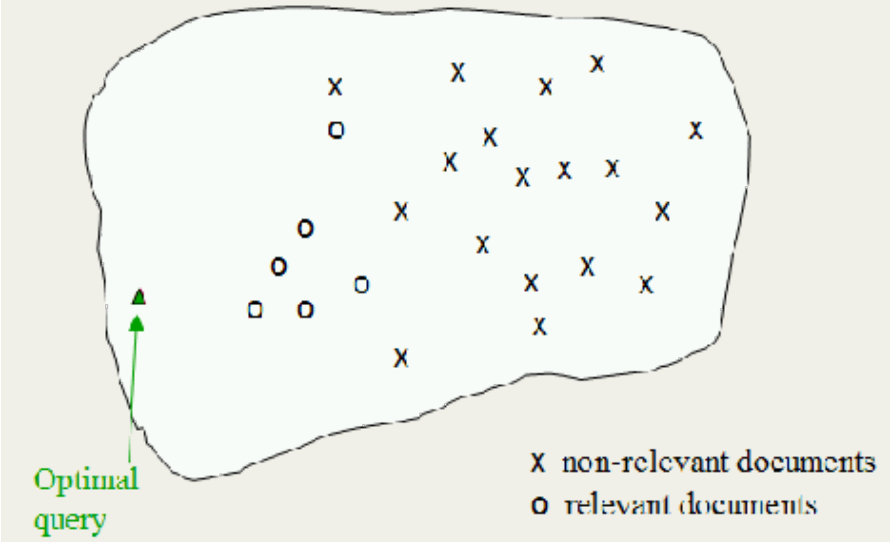


Optimal Query

- Assume that the relevant set of image C_r are known and N is the total number of images
- Then the best query that ranks all and only the relevant queries at the top is:

$$\vec{q}_{opt} = \arg\max_{\vec{q}} \left(sim(\vec{q}, C_r) - sim(\vec{q}, C_{nr}) \right)$$

$$\vec{q}_{opt} = \frac{1}{|C_r|} \sum_{\forall \vec{d}_j \in C_r} \overrightarrow{d_j} - \frac{1}{N-|C_r|} \sum_{\forall \vec{d}_j \notin C_r} \overrightarrow{d_j}$$



Standard Rochio Method

- Since all relevant images unknown, just use the known relevant (D_r) and irrelevant (D_n) sets of images and include the initial query q

$$\vec{q}_m = \alpha \vec{q} + \frac{\beta}{|D_r|} \sum_{\forall \vec{d_j} \in D_r} \vec{d_j} - \frac{\gamma}{|D_n|} \sum_{\forall \vec{d_j} \in D_n} \vec{d_j}$$

- α: Tunable weight for initial query
- β: Tunable weight for relevant documents
- γ: Tunable weight for irrelevant documents
- Recommended parameters: $(\alpha, \beta, \gamma) = (1, 0.75, 0.25)$
- If only positive feedback: $\gamma = 0$

Ide Regular Method

- Since more feedback should perhaps increase the degree of reformulation, do not normalize for amount of feedback

$$\vec{q}_m = \alpha\vec{q} + \beta \sum_{\forall \vec{d_j} \in D_r} \vec{d_j} - \gamma \sum_{\forall \vec{d_j} \in D_n} \vec{d_j}$$

- α: Tunable weight for initial query
- β: Tunable weight for relevant documents
- γ: Tunable weight for irrelevant documents

Ide "Dec Hi" Method

- Bias towards rejecting **just** the highest ranked of the irrelevant documents (use only marked nonrelevant image which received the highest ranking from the IR system as negative feedback, $|D_n| = 1$)

$$\vec{q}_m = \alpha\vec{q} + \beta \sum_{\forall \overrightarrow{d_j} \in D_r} \overrightarrow{d_j} - \gamma \max_{non-relevant}\left(\overrightarrow{d_j}\right)$$

- α: Tunable weight for initial query
- β: Tunable weight for relevant documents
- γ: Tunable weight for irrelevant documents

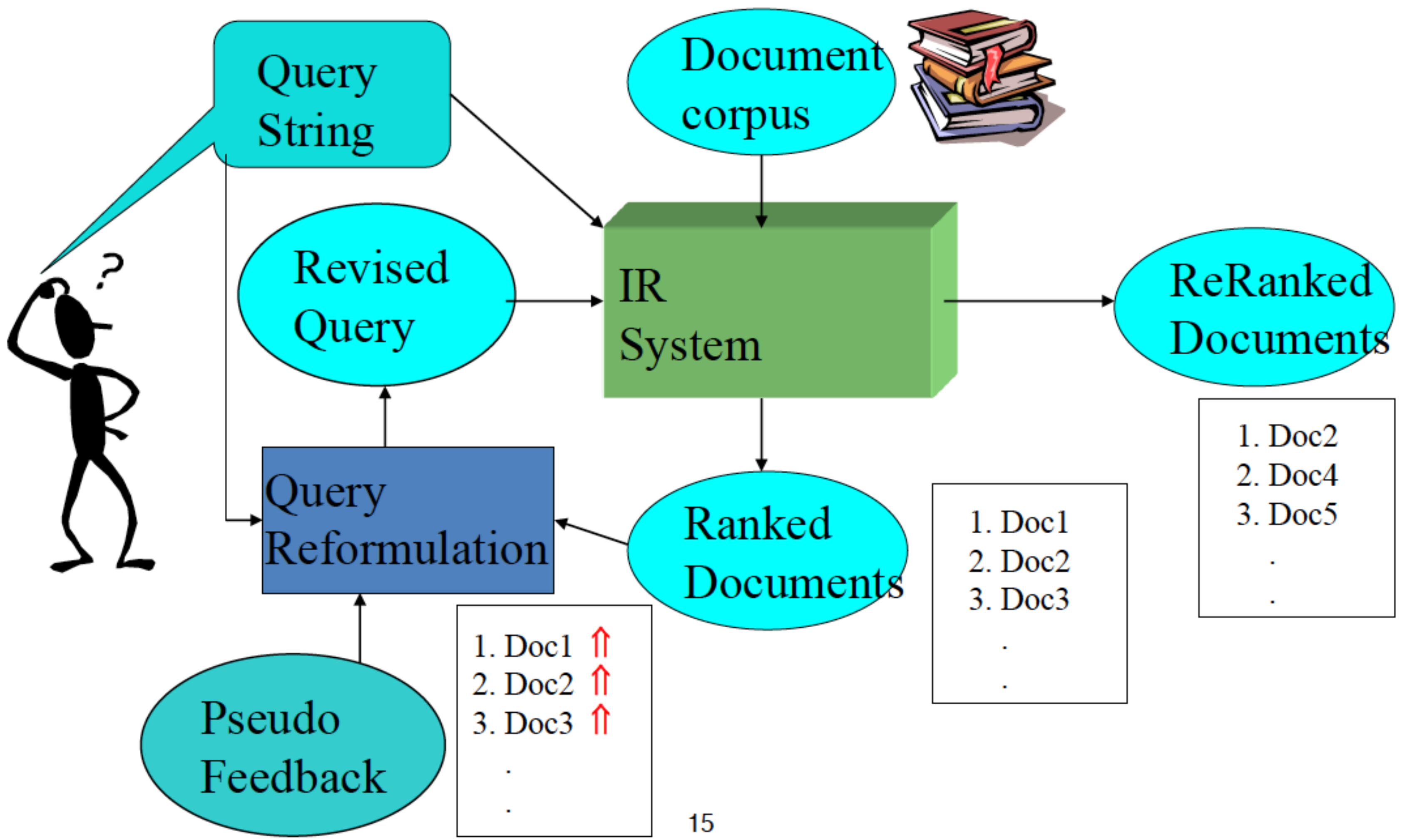
Why is Feedback Not Widely Used

- Users sometimes reluctant to provide explicit feedback.
- Results in long queries that require more computation to retrieve, and search engines process lots of queries and allow little time for each one.
- Makes it harder to understand why a particular image was retrieved.

Pseudo/Blind Feedback

- Use relevance feedback methods without explicit user input.
- Just assume the top m retrieved images are relevant and use them to reformulate the query.
- Allows for query expansion that includes terms that are correlated with the query terms.

Pseudo/Blind Feedback



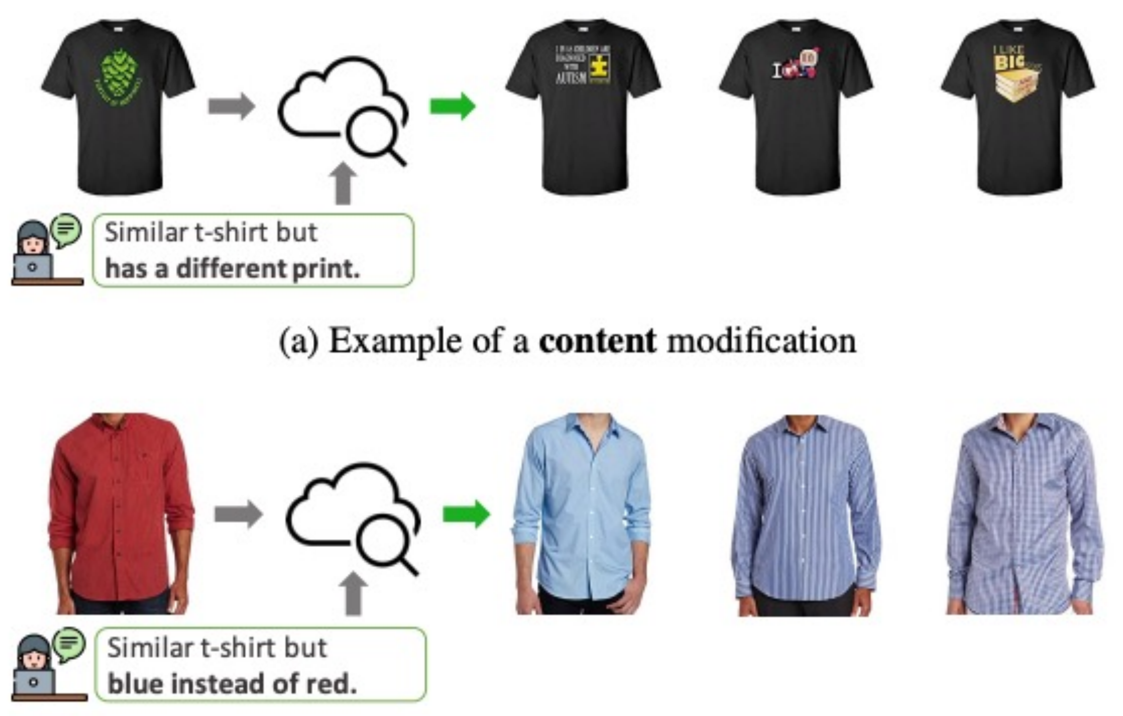
Pseudo/Blind Feedback

- Found to improve performance on TREC competition ad-hoc retrieval task.
- Works even better if top documents must also satisfy additional Boolean constraints in order to be used in feedback.
  - If the query is about copper mines and the top several documents are all about mines in Chile, then there may be query drift in the direction of documents on Chile

Image Retrieval with Text Feedback

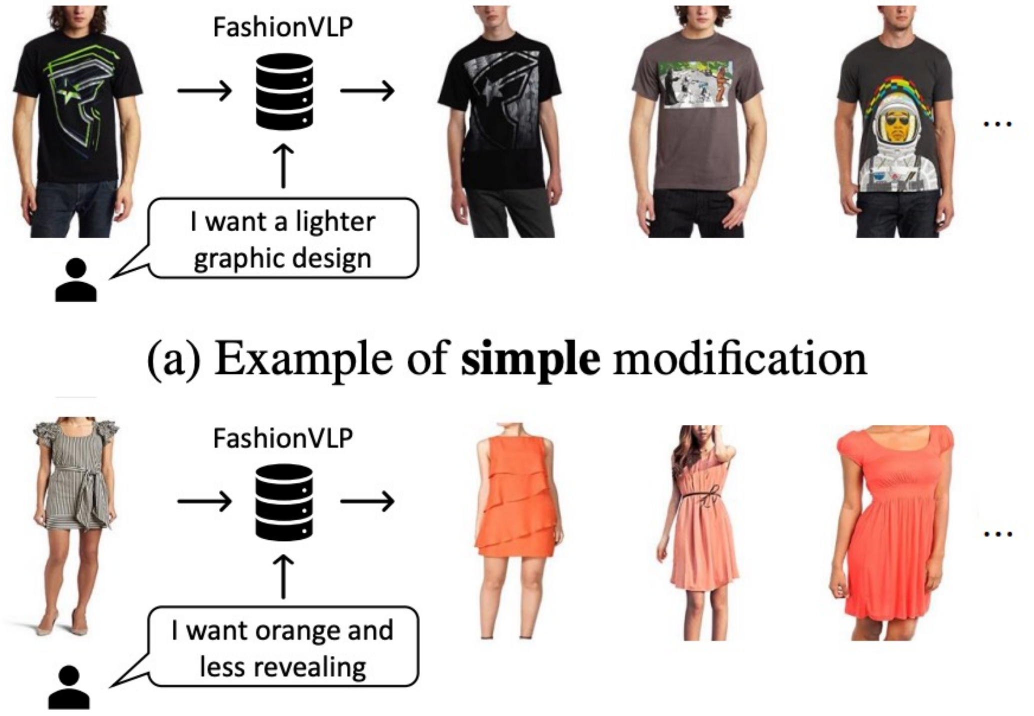
- Feedback in natural language

Fashion Image Retrieval with Text Feedback



(a) Example of a **content** modification

(b) Example of a **style** modification



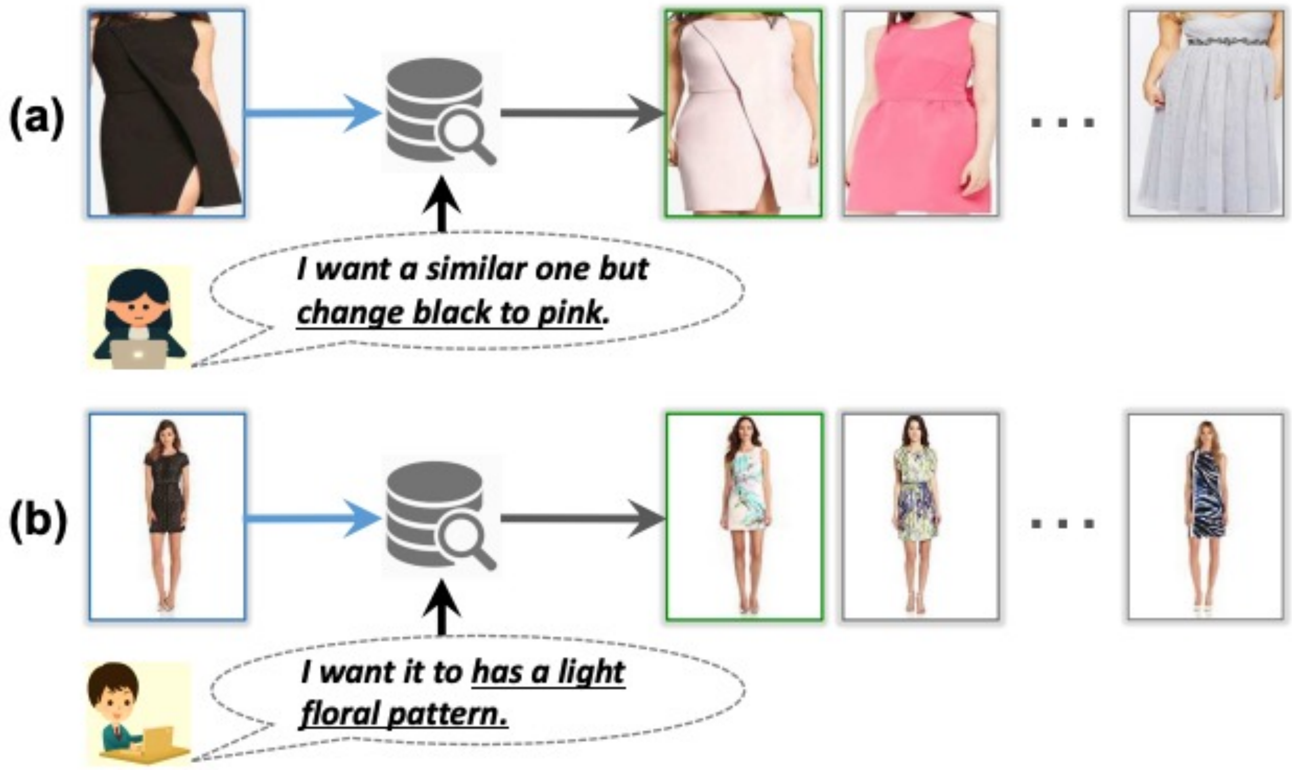
(a) Example of **simple** modification

(b) Example of **compound** modification

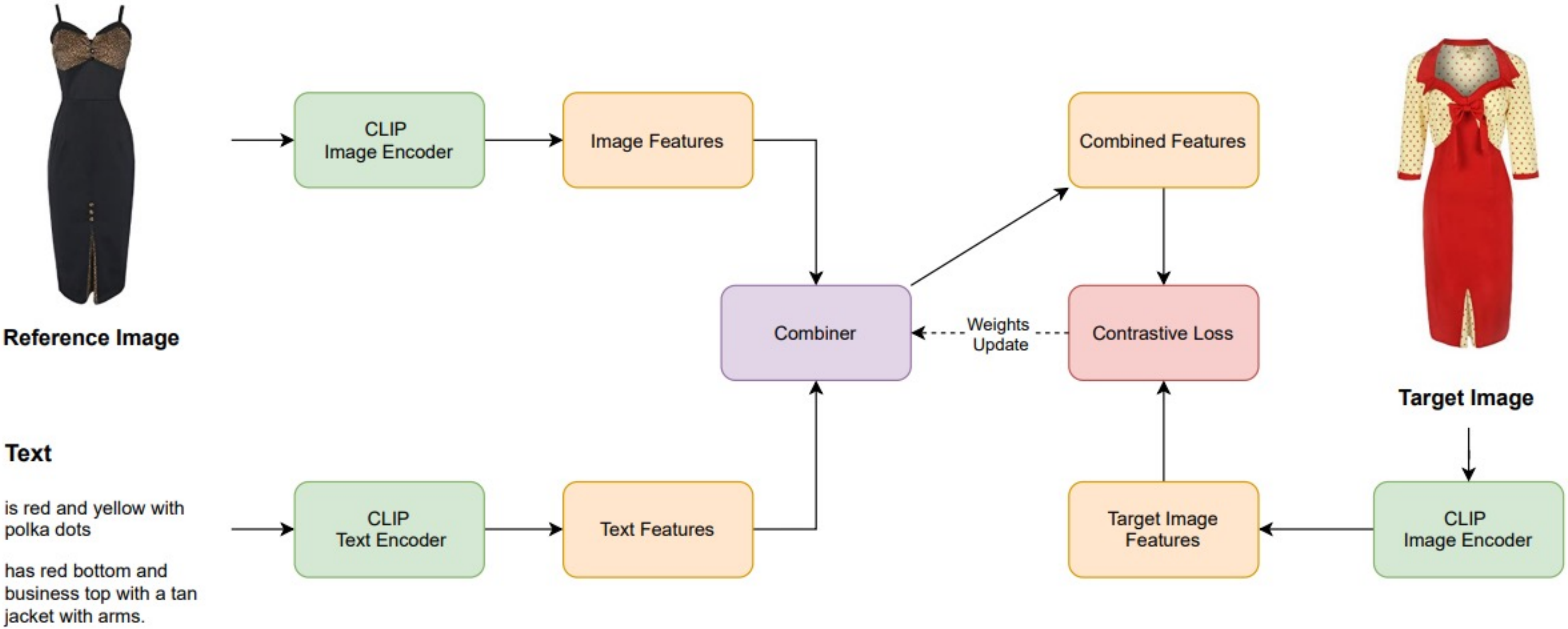
Lee et al., “CoSMo: Content-Style Modulation for Image Retrieval with Text Feedback”, CVPR 2021
Sonam et al., “FashionVLP: Vision Language Transformer for Fashion Retrieval with Feedback”, CVPR 2022

Fashion Image Retrieval

- Pre-trained CLIP features alone may not work here
- Train new vision-language encoders

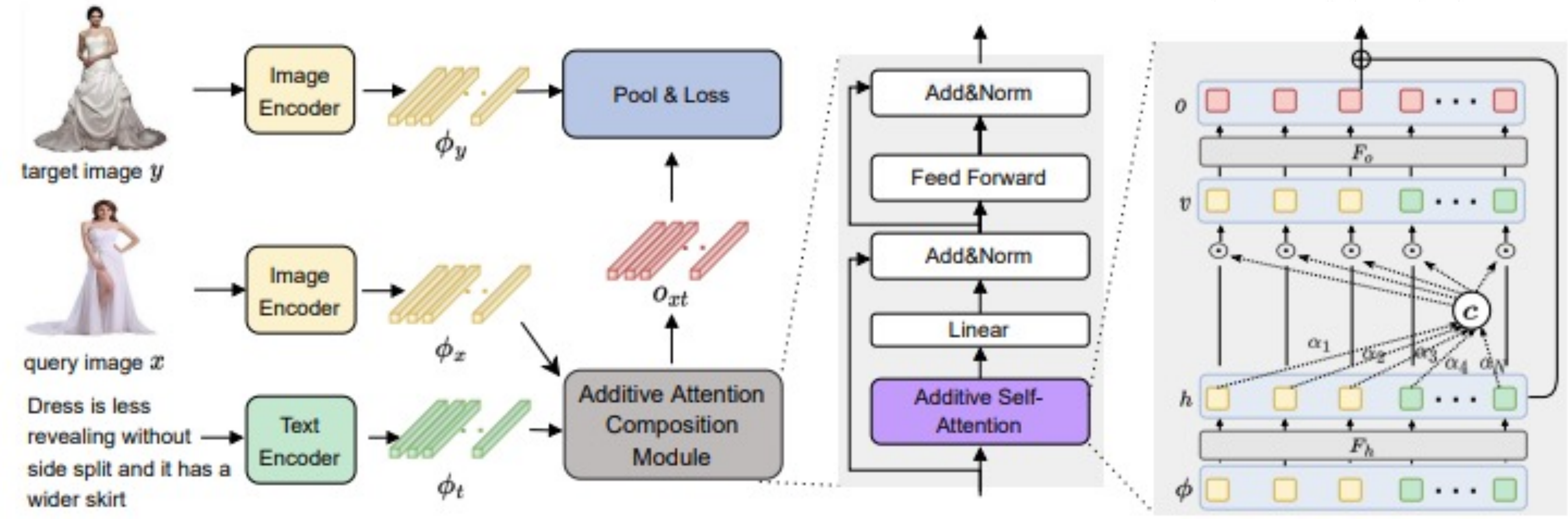


Fashion Image Retrieval



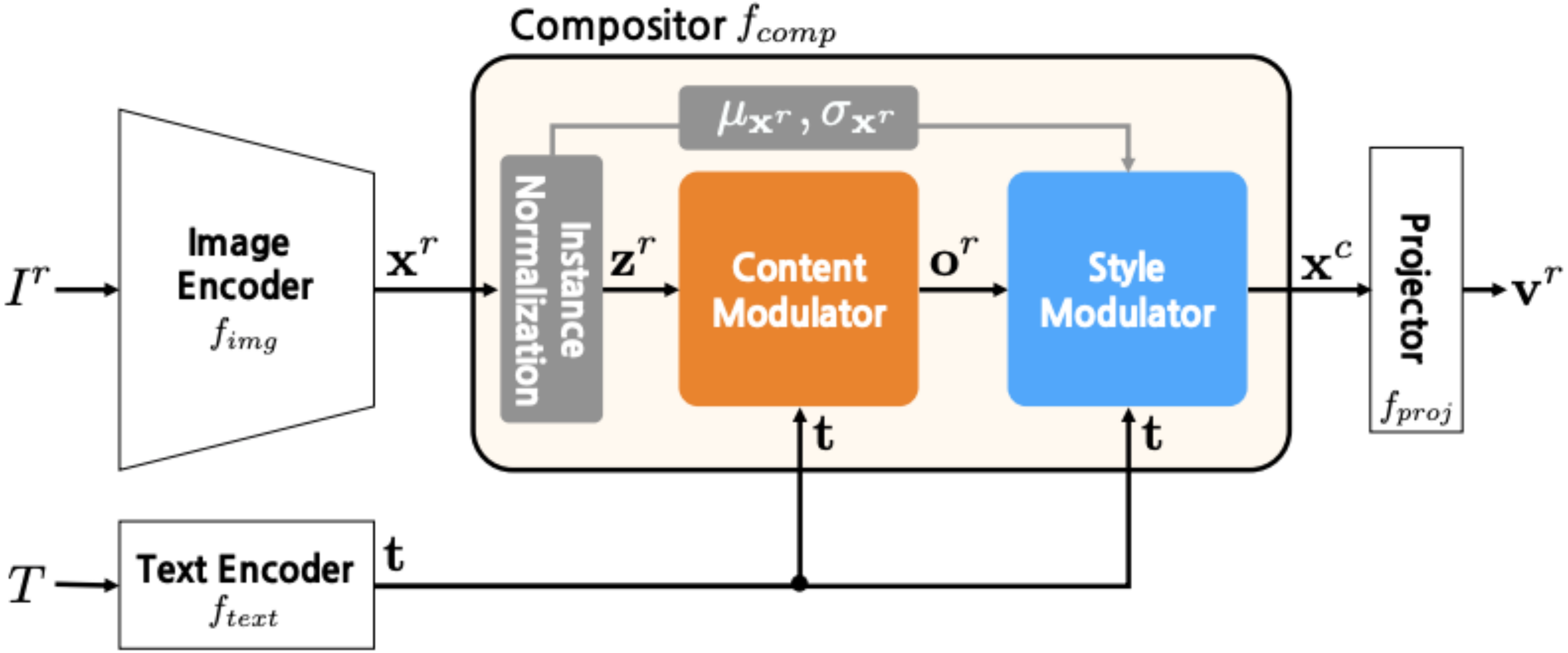
Alberto et al., “Conditioned Image Retrieval for Fashion using Contrastive Learning and CLIP-based Features”, MM Asia 2021

Fashion Image Retrieval



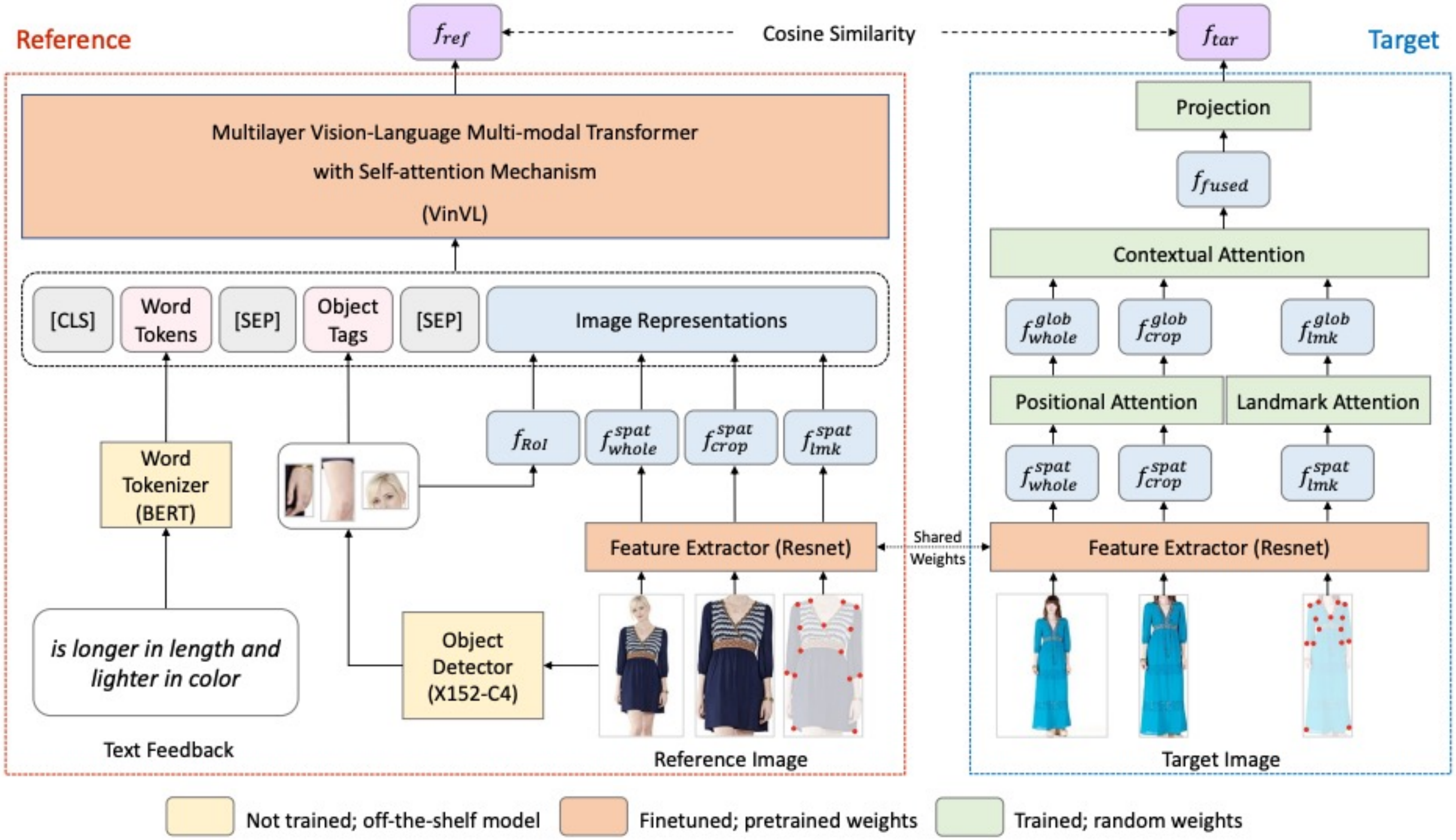
Tian et al., "Fashion Image Retrieval with Text Feedback by Additive Attention Compositional Learning", WACV 2023

Fashion Image Retrieval



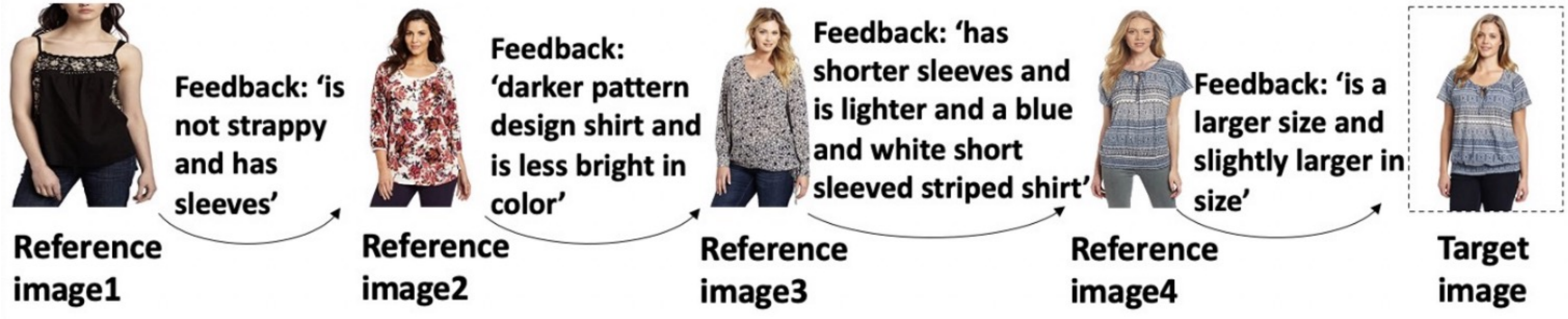
Lee et al., "CoSMo: Content-Style Modulation for Image Retrieval with Text Feedback", CVPR 2021

Fashion Image Retrieval



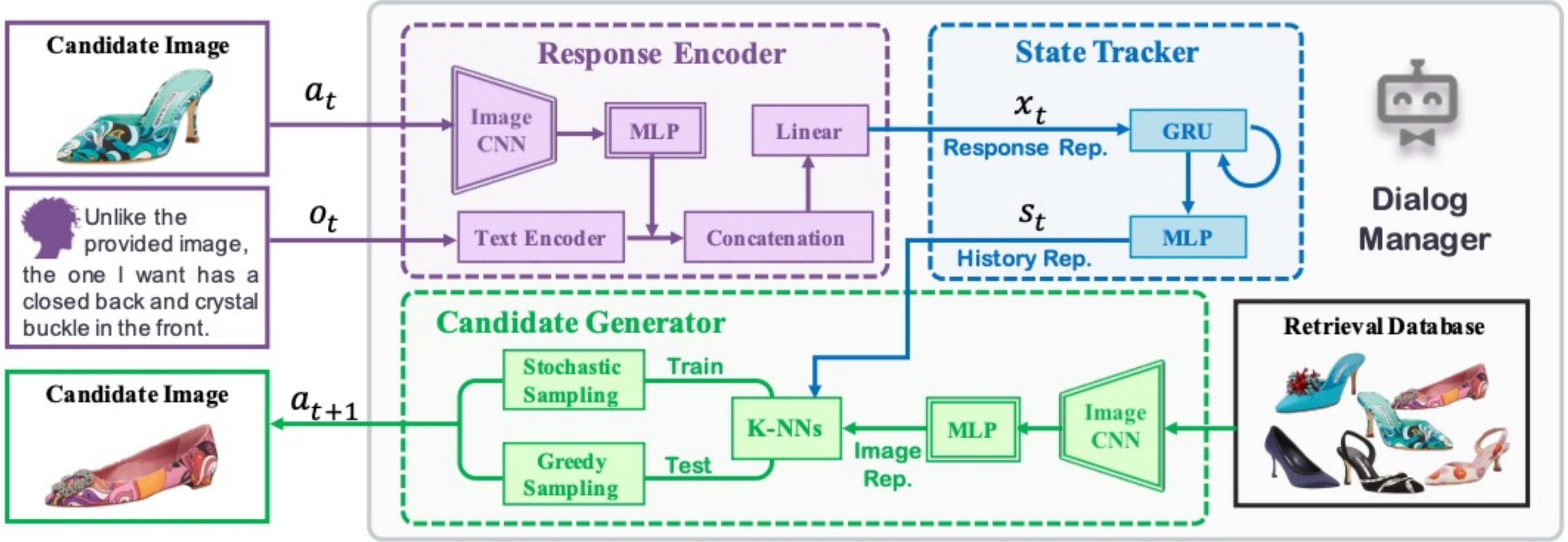
Sonam et al., "FashionVLP: Vision Language Transformer for Fashion Retrieval with Feedback", CVPR 2022

Interactive Image Retrieval



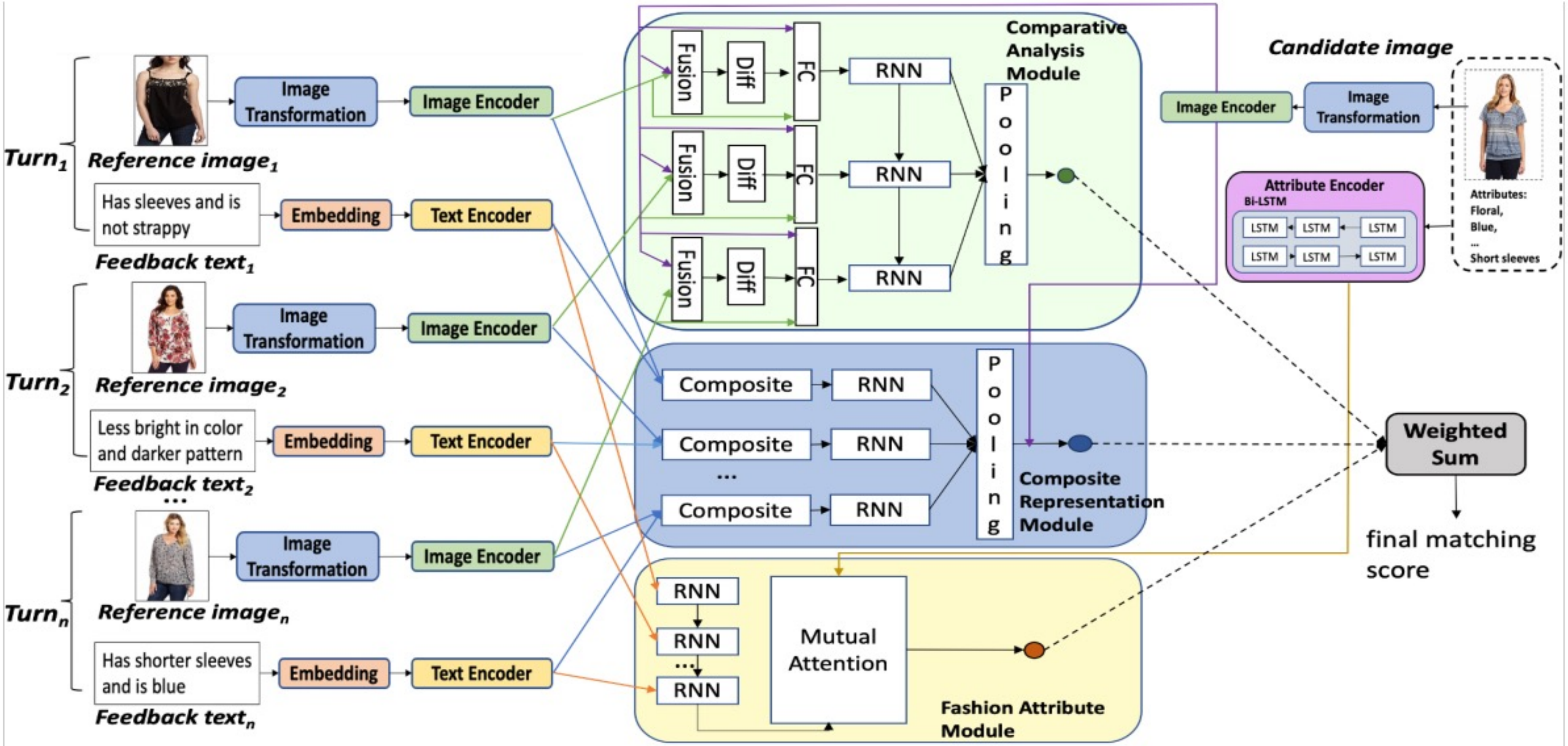
Interactive Image Retrieval

- Agent incorporates the user's feedback (natural language) to iteratively refine retrieval results



Xiaoxiao et al., "Dialog-based Interactive Image Retrieval", NeurIPS 2018

Interactive Image Retrieval



Yuan et al., "Conversational Fashion Image Retrieval via Multiturn Natural Language Feedback", SIGIR 2021